
ROADASSIST BHARAT

Team Charter

Engineering Constitution

| *One Platform. Every Vehicle. Every Phone. Every Road.*

Document docs/00-team-charter.md

Version 1.0 — planning baseline

Team 4 engineers · 22 weeks · 11 sprints · 18 phases

Status Baseline (frozen at Sprint 0 exit)

RoadAssist Bharat — Team Charter & Engineering Constitution

"One Platform. Every Vehicle. Every Phone. Every Road." Document owner: CTO · Version 1.0 · Status: Baseline (frozen at Sprint 0 exit)

1. Team Composition

Four engineers. Every person is a **Lead** — meaning they own an architectural vertical end-to-end, not a ticket queue. No one is a "helper."

ID	Role	Owns	Secondary (backup)
D1	Backend & Data Lead	All Node/NestJS services, PostgreSQL/PostGIS schema, Kafka event bus, API contracts, payments	Backs up D3 on AI-gateway plumbing
D2	Frontend & Mobile Lead	React Native app, Next.js portals (Gov / Mechanic / Admin), design system, offline UI, Android Auto	Backs up D4 on E2E test authoring
D3	AI & Data Science Lead	All 9 AI subsystems, model training/serving, on-device TFLite, telemetry feature store, IVR/NLU	Backs up D1 on analytics service
D4	DevOps, QA & Security Lead	Kubernetes, Terraform, CI/CD, observability, load/security testing, SMS/IVR gateway ops, compliance	Backs up D2 on portal builds

Rotating hats (1 sprint each, round-robin): - **Release Captain** — owns the sprint's release train, cuts the tag, writes release notes. - **Incident Commander** — on-call for staging/prod, runs postmortems. - **Scribe** — keeps ADRs and the decision log current.

Named slots (fill in):

D1 = _____ D2 = _____

D3 = _____ D4 = _____

2. Ownership Matrix (RACI)

Area	D1	D2	D3	D4
System architecture / ADRs	A	C	C	C
Database schema & migrations	A/R	I	C	C
API contract (OpenAPI/proto)	A/R	C	C	I
Auth, RBAC, token lifecycle	A/R	C	I	C
Booking / Dispatch engine	A/R	C	C	I

Area	D1	D2	D3	D4
Mobile app (RN)	C	A/R	I	C
Web portals (Next.js)	C	A/R	I	C
Design system & a11y	I	A/R	I	I
Offline DB + sync client	C	A/R	C	I
Sync conflict resolution (server)	A/R	C	I	I
AI models & training	I	I	A/R	C
On-device inference (TFLite/ONNX)	I	C	A/R	I
Voice assistant / IVR NLU	I	C	A/R	C
Maps, routing, geofencing	C	C	C	A/R shared with D1
Emergency (112 / SOS) engine	A/R	C	C	C
Government dashboard	C	A/R	C	I
Analytics / ClickHouse	C	I	A/R	C
Kubernetes, Terraform, CI/CD	I	I	I	A/R
Observability & SLOs	C	C	C	A/R
Security testing & pentest	C	C	C	A/R
Load & chaos testing	C	I	C	A/R
DPDP / compliance evidence	C	I	C	A/R
Pitch deck / business model	C	C	C	C — A = CTO

A = Accountable · R = Responsible · C = Consulted · I = Informed

3. Cadence

Daily

- **09:45 IST — Standup (12 min, hard stop).** Format below. Async in Slack [#ra-standup](#) if anyone is travelling.
- **Focus block 10:00–13:00.** No meetings, no PR review, notifications off.
- **16:30 — PR sweep.** Every open PR must be reviewed or explicitly deferred with a reason.
- **18:00 — Green build gate.** [main](#) must be green at EOD. Whoever broke it fixes it or reverts it.

Standup format (each person, 3 lines max):

```

Yesterday : <shipped thing + PR link>
Today      : <one primary outcome>
Blocked    : <blocker + who unblocks it, or "none">
```

Discussion is deferred to a "parking lot" after the 12 minutes with only the affected people.

Weekly

Day	Ritual	Duration	Who
Mon	Sprint planning (odd weeks) / Mid-sprint checkpoint (even weeks)	60 min	All
Tue	Architecture review — ADRs raised last week	45 min	All
Wed	Design review — Figma → build parity	30 min	D2 + 1
Thu	Risk & security review — new threats, dependency CVEs	30 min	D4 + all
Fri	Demo (recorded, 20 min) + Retro (25 min)	45 min	All

Sprint

2 weeks. 10 working days. Capacity per person per sprint: **8 ideal days** (2 days reserved for review, support, and unplanned work). Team velocity target: **32 ideal days/sprint**.

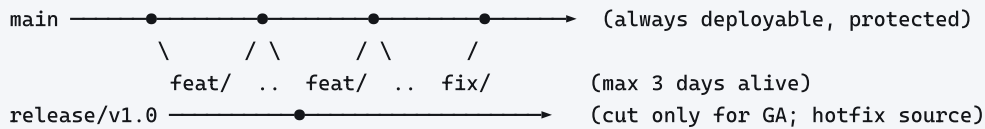
Sprint ends with: 1. A recorded demo of working software (never slides). 2. A tagged release `v0.<sprint>.0` deployed to staging. 3. A retro with at most **2 action items**, each with an owner and a due date.

4. Sprint Calendar & Milestones

Sprint	Weeks	Phases covered	Milestone
S0	1–2	P0 Research, P1 Planning	M0 — Baseline frozen: PRD, ADR-000..010, repo scaffolded, CI green
S1	3–4	P2 Architecture, P3 Database	M1 — Skeleton alive: all services boot, migrations run, ER diagram signed off
S2	5–6	P4 Auth, P5 Backend APIs (I)	M2 — Login works: OTP → JWT → RBAC across 3 client types
S3	7–8	P5 Backend APIs (II), P6 Frontend (I)	M3 — First booking: end-to-end request → mechanic accept → complete
S4	9–10	P6 Frontend (II), P7 AI (I)	M4 — Vertical slice: mobile app + live dispatch on a map
S5	11–12	P7 AI (II), P8 Offline Engine	M5 — Works with no internet: airplane-mode booking that syncs
S6	13–14	P9 Maps, P10 Vehicle Diagnostics	M6 — Diagnose & route: OBD/photo diagnosis + offline routing
S7	15–16	P11 Emergency, P12 Gov Dashboard	M7 — Crash to responder: auto-detect → SOS → 112 handoff → gov map
S8	17–18	P13 Analytics, P14 Testing	M8 — Quality bar: ≥80% coverage, load test 10k RPS passed
S9	19–20	P15 Deployment, P16 Optimization	M9 — Production: multi-AZ prod, DR drill passed, p95 < 300 ms
S10	21–22	P17 Future Roadmap + GA hardening	M10 — GA + pitch: public beta, pitch deck, pilot MoU draft

5. Git Workflow

Branching model — Trunk-based with short-lived branches



- `main` is protected. No direct pushes. Squash-merge only.
- Branch lifetime $\leq$ **3 days**. If it lives longer, it was scoped wrong — split it.
- Feature flags (`unleash`) over long-lived branches. Always.
- `release/vX.Y` cut at GA. Hotfixes branch from it and cherry-pick back to `main`.

Branch naming

<type>/<phase>-<owner>-<short-kebab-summary>

```
feat/p05-d1-booking-state-machine
fix/p08-d2-sync-conflict-toast
chore/p15-d4-argocd-bootstrap
model/p07-d3-crash-detect-v2
docs/p02-d1-adr-012-event-bus
spike/p10-d3-obd-ble-feasibility
```

Types: `feat` `fix` `chore` `refactor` `perf` `test` `docs` `model` `spike` `hotfix`

Commit naming — Conventional Commits, enforced by commitlint

<type>(<scope>): <imperative summary, $\leq$ 72 chars>

<body: why, not what — wrap at 100>

Refs: RA-142

BREAKING CHANGE: booking.status enum renamed PENDING→REQUESTED

Scopes: `auth` `booking` `dispatch` `vehicle` `maps` `emergency` `notify` `sync` `analytics` `gov` `ai` `mobile` `web`
`infra` `db`

Good:

```
feat(dispatch): rank mechanics by ETA-adjusted acceptance probability
fix(sync): drop tombstoned rows before applying remote patch
```

Bad: `update code`, `fixes`, `WIP`, `final final`

6. Pull Request Rules

Hard rules — CI blocks the merge: 1. PR diff $\leq$ **400 changed lines** (excluding lockfiles, generated code, snapshots). Larger needs an explicit `size/xl` label plus CTO approval. 2. **1 approving review minimum, 2 for:** DB migrations, auth/crypto, payment flows, emergency service, anything touching PII. 3. The reviewer must **not** be the author's backup for that area on trivial PRs — rotate reviewers so knowledge spreads. 4. All CI gates green: lint, typecheck, unit, integration, coverage delta ≥ 0 , SAST, dependency audit, license check, container scan. 5. PR description uses the template. An empty template body = automatic request-changes. 6. Migrations must include a tested **down** path or an explicit "forward-only, approved by D1" note. 7. No `TODO` without a ticket ID. No commented-out code. No `console.log` in shipped paths. 8. Review SLA: **first response within 4 working hours**. If you can't, reassign — don't sit on it.

PR template:

```
## What
One paragraph. What changed, in user terms.

## Why
Link the ticket / ADR. What problem does this solve?

## How
Notable design decisions and anything a reviewer would otherwise have to reverse-engineer.

## Risk
- Blast radius: [ ] single service [ ] cross-service [ ] data migration [ ] auth/PII
- Rollback plan:
- Feature flag: `flag.name` (default OFF)

## Verification
- [ ] Unit tests added/updated
- [ ] Integration test covers the happy path + 1 failure path
- [ ] Manually verified on: [ ] Android [ ] iOS [ ] Web [ ] low-end device [ ] airplane mode
- [ ] Screenshots / recording attached (UI changes)
- [ ] Load impact considered (new N+1? new hot query? index present?)

## Docs
- [ ] OpenAPI updated [ ] ADR written [ ] Runbook updated [ ] README updated [ ] N/A + reason
```

7. Code Review Checklist

The reviewer works this list. It's the contract — no vibes-based reviews.

Correctness - [] Does it actually do what the PR says? Read the test, not just the code. - [] Edge cases: empty list, null, zero, negative, duplicate, concurrent, timezone, `NULL` vs missing. - [] Error paths: what happens on network failure, DB timeout, partial write? - [] Idempotency: can this be safely retried? (All write APIs must be — `Idempotency-Key` header.)

Data - [] Migration is backwards-compatible with the currently-deployed code (expand → migrate → contract). - [] New query has a supporting index; `EXPLAIN ANALYZE` attached for anything on a hot path. - [] Soft delete respected (`deleted_at IS NULL` in every read path). - [] Audit log entry written for anything state-changing on a regulated entity.

Security - [] AuthZ check present at the resource level, not just the route level (no IDOR). - [] Input validated with a schema (Zod / class-validator) at the boundary. - [] No raw string SQL concatenation. Parameterized or query-builder only. - [] No secrets, keys, tokens, or PII in code, logs, or error messages. - [] Rate limit applied to any unauthenticated or

expensive endpoint.

Performance - ☐ No N+1. No unbounded query (every list endpoint is paginated, max page size enforced). - ☐ Payload size sane for a 2G connection — is this shipping fields the client never reads? - ☐ Caching considered; cache invalidation path is explicit.

Frontend-specific - ☐ Works offline or degrades honestly (never a silent failure — always a visible state). - ☐ Loading / empty / error / offline states all implemented. - ☐ Touch targets $\geq 44 \times 44$ dp; contrast $\geq 4.5:1$; screen-reader labels present. - ☐ Tested on a 2 GB RAM Android device profile, not just the simulator. - ☐ i18n: no hardcoded user-facing strings.

AI-specific - ☐ Model version pinned; model card updated; eval metrics reported vs. previous version. - ☐ Fallback path when the model is unavailable or confidence is below threshold. - ☐ No PII in training data or feature store without a documented lawful basis. - ☐ Latency and model size budget respected (on-device: ≤ 15 MB, ≤ 120 ms p95).

Craft - ☐ Naming reads like the domain, not like the implementation. - ☐ The code matches the conventions of the file it lives in. - ☐ Comments explain *why*, and only where the why isn't obvious. - ☐ No speculative generality — no abstraction with exactly one caller.

8. Definition of Done

A story is Done only when **all** of these are true. Not "done except tests."

1. Acceptance criteria in the ticket are all demonstrably met.
2. Code merged to `main` via a reviewed PR.
3. Unit tests written; branch coverage on new code $\geq 80\%$.
4. At least one integration test covers the happy path and one failure path.
5. OpenAPI / proto contract updated and the generated client regenerated.
6. Feature flag defined, default OFF, and the ON path tested.
7. Observability: at least one metric, one structured log event, and a trace span exist for the new path.
8. Alert defined if the feature has an SLO.
9. Docs updated: API reference, and the runbook if it can page someone.
10. Deployed to staging and smoke-tested there.
11. Works on the **low-end reference device** (Android 10, 2 GB RAM, 3G throttled) if user-facing.
12. Accessibility pass (WCAG 2.1 AA) if user-facing.
13. Security checklist signed off by D4 if it touches auth, PII, payments, or emergency.
14. Demoed in the Friday demo.

Definition of Ready (before a story enters a sprint): has acceptance criteria, has a design/API contract, has no unresolved external dependency, is estimated, and is ≤ 3 ideal days.

9. Documentation Standards

```
docs/
├── 00-team-charter.md           ← this file
├── 01-master-roadmap.md        ← all 18 phases, cross-team
├── 02-backend-lead-roadmap.md  ← D1
├── 03-frontend-lead-roadmap.md ← D2
├── 04-ai-lead-roadmap.md       ← D3
├── 05-devops-qa-lead-roadmap.md ← D4
├── adr/                        ← Architecture Decision Records, numbered, immutable
│   └── 0012-event-bus-redpanda-over-kafka.md
├── api/                       ← generated OpenAPI + proto, never hand-edited
├── runbooks/                  ← one per alert; every alert links to its runbook
├── models/                    ← one model card per deployed model
└── decisions.md               ← running decision log (date, decision, who, why)
```

Rules - Every non-obvious technical choice gets an **ADR**. Format: Context → Options considered → Decision → Consequences → Status. ADRs are never edited after acceptance; they're superseded by a new one. - Every alert has a **runbook** with: what it means, how to confirm, how to mitigate, how to escalate. An alert without a runbook gets deleted. - Every deployed model has a **model card**: purpose, training data + provenance, metrics, known failure modes, bias evaluation, fallback behaviour, owner. - Diagrams are **Mermaid in markdown**, not images. Diagrams that can't be diffed rot. - READMEs answer exactly four questions: what is this, how do I run it, how do I test it, who owns it. - Docs are updated **in the same PR** as the code. A follow-up docs PR never happens.

10. Communication Protocol

Channel	Use for	Response expectation
#ra-standup	Async standups	Post by 09:45
#ra-dev	Technical discussion	Best effort
#ra-prs	PR notifications	4 working hours
#ra-alerts	Automated alerts only, no chat	On-call: 15 min ack
#ra-incidents	Live incident coordination	Immediate
ADR + PR	Any decision anyone will ask "why?" about in 3 months	2 working days

Escalation: blocked > 2 hours → say so in #ra-dev . Blocked > 4 hours → escalate to CTO. Sitting quietly on a blocker is the only genuinely unacceptable behaviour on this team.

11. Quality Gates (enforced in CI, not by goodwill)

Gate	Threshold	Blocks
ESLint / Ruff	0 errors	PR

Gate	Threshold	Blocks
TypeScript / mypy	strict, 0 errors	PR
Unit coverage (new code)	≥ 80% branch	PR
Integration suite	100% pass	PR
SAST (Semgrep, CodeQL)	0 high/critical	PR
Dependency audit (osv-scanner)	0 critical, high needs waiver	PR
Container scan (Trivy)	0 high/critical in final image	Release
License check	no GPL/AGPL in distributed code	PR
API contract diff	no breaking change without version bump	PR
Bundle size (web)	≤ 180 KB gz initial route	PR
APK size	≤ 28 MB base	Release
Load test	p95 < 300 ms @ 10k RPS	Release
Lighthouse (portals)	Perf ≥ 90, A11y ≥ 95	Release
DAST (ZAP)	0 high	Release

12. Risk Register (top 10, reviewed every Thursday)

#	Risk	Impact	Prob	Owner	Mitigation
R1	Offline sync conflicts corrupt booking state	Critical	Med	D1+D2	CRDT-style LWW per-field + server-authoritative state machine; conflict test suite from S5
R2	SMS/IVR gateway DLT approval delays (TRAI)	High	High	D4	Start DLT template registration in S0; keep 2 gateway vendors integrated
R3	Crash-detection false positives dispatch real 112 units	Critical	Med	D3	Two-stage confirm (sensor + 30s user cancel window + call-back); never auto-dispatch on model alone
R4	Map/routing licensing cost at national scale	High	Med	D4	OSM + self-hosted Valhalla/MapLibre as the default; commercial tiles only as premium fallback
R5	DPDP Act non-compliance (consent, localization)	Critical	Med	D4	Consent ledger from P4; all data in-India regions; DPIA before beta
R6	On-device AI too heavy for 2 GB devices	High	Med	D3	Hard budget: ≤15 MB, INT8 quantized, ONNX/TFLite; server fallback
R7	Mechanic supply density too low for ETA promises	High	High	PM/D1	Dispatch honestly surfaces "no mechanic within X"; pre-seed partner network per pilot city
R8	4-person team under-capacity for 18 phases	High	High	CTO	Phases 12/13/17 are deliberately thin-sliced; feature flags let scope drop without branch surgery

#	Risk	Impact	Prob	Owner	Mitigation
R9	Emergency service outage during a real emergency	Critical	Low	D4	Multi-AZ + degraded SMS-only path that works with the whole app stack down
R10	Key-person dependency (single lead per vertical)	High	Med	All	Mandatory backup owner (\$1), rotating reviewers, ADRs, no undocumented tribal knowledge

13. Non-Negotiables

1. **Emergency paths never regress.** Any change touching SOS/112 needs 2 approvals, a dedicated test run, and a staged rollout.
2. **Offline is a first-class mode, not a fallback.** If a feature can't degrade, it needs an explicit, visible offline state.
3. **Feature phones are users.** Every core journey (request help, track, cancel, confirm) must be completable over SMS/IVR.
4. **No PII leaves India.** Ever. Including logs, backups, error reports, and third-party analytics.
5. **We never fake it in a demo.** If it's mocked, we say it's mocked, on the slide.